

WEENUKA RAJAPAKSHA

Computer Science & Engineering Undergraduate | University of Moratuwa

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PROFILE

Third-year Computer Science & Engineering undergraduate at the University of Moratuwa with hands-on experience across AI/ML, computer vision, full-stack web and mobile development, and embedded/IoT systems. Passionate about turning ideas into functional, real-world applications, with a strong focus on clean code, problem-solving, and continuous learning. Comfortable working independently and in teams, with a strong record of delivering working prototypes from concept to demo.

EDUCATION

B.Sc. Engineering (Hons) in Computer Science & Engineering 2024 – Present

University of Moratuwa, Sri Lanka — Currently in 3rd Year

G.C.E. Advanced Level — Physics, Combined Mathematics, Chemistry (3 A's) 2022

Richmond College, Galle — Z-Score: 2.502

G.C.E. Ordinary Level — 9 A's, including Mathematics, Science, English, and ICT

TECHNICAL SKILLS

Programming	Python, C, C++, Java, JavaScript, Dart
Web & Mobile	React, Flutter, Flask, Node.js, Express.js, REST APIs, HTML/CSS, Tailwind CSS
AI / ML / CV	Scikit-learn, Pandas, NumPy, OpenCV, MediaPipe, YOLO, EasyOCR, Feature Engineering, Librosa
Embedded / IoT	ESP32, Arduino, VHDL (Xilinx Vivado, Basys 3 FPGA), Embedded C
Databases	MySQL, PostgreSQL, SQLite, MongoDB
Tools & Platforms	Git, VS Code, Google Gemini API, Figma, Ubuntu
Languages	English (Fluent), Sinhala (Native), Tamil (Basic reading and writing)

TECHNICAL PROJECTS

Sinhala Office Assist (සිංහල කාර්යාල සහායක) 2026 — In Progress

Flutter mobile app that records Sinhala speech and uses the Gemini API to automatically generate formal Sinhala office documents.

Technologies: Flutter, Dart, Gemini API

SoundScout AI — Event Infrastructure & Vendor-Bidding Marketplace 2026

AI-powered marketplace connecting event organizers with vendors through a bidding system, built for the Algostrom Idealize Ideathon with a custom design system and branding.

Technologies: React, Vite, Tailwind CSS, Node.js, Express.js, PostgreSQL, Gemini API

FeverGuard AI — Intelligent Fever Monitoring System 2025

AI-powered healthcare solution combining an IoT wearable and web app for continuous body temperature monitoring; auto-generates fever charts, analyzes fever patterns with a custom ML model, and recommends relevant medical tests and specialists.

Technologies: ESP32-C6, DS18B20, OLED Display, Python, Flask, Machine Learning, HTML/CSS, JavaScript

Smart Traffic Violation Detection System 2025

Full-stack computer vision application that detects traffic violations, identifies vehicles, and reads license plates via OCR, with a web dashboard for managing detected offenses.

Technologies: Python, YOLO, OpenCV, EasyOCR, Flask, SQLite

Virtual Air Drum Kit 2025

Computer vision-based virtual drum kit using real-time hand tracking to detect drum hits and play corresponding sounds through a webcam, with no additional hardware required.

Technologies: Python, OpenCV, MediaPipe, Pygame

AI Guitar Assistant 2025

AI-powered music analysis app that processes uploaded audio to detect key, tempo, and chords, generating guitar-friendly chord sheets for practice.

Technologies: Python, Flask, Librosa, SQLite

YouTube Viewership Forecasting	2026 — <i>In Progress</i>
Machine learning project predicting YouTube video viewership for the Sri Lankan audience; involves custom dataset collection, feature engineering, and predictive modeling.	
Technologies: Python, YouTube Data API v3, Pandas, Scikit-learn	
Song Popularity Prediction — Kaggle MLX 2.0 Competition	2025
Built regression models to predict song popularity, applying data preprocessing, feature engineering, cross-validation, and model comparison to improve accuracy.	
Technologies: Python, Scikit-learn, Pandas, NumPy	
NFC-Based Vehicle Mileage Management System	2024
Embedded system using NFC technology to identify vehicles and automatically record mileage, simplifying vehicle tracking and improving record accuracy.	
Technologies: Arduino, NFC Module, Embedded C	
Personal Portfolio Website	2026
Designed and developed a premium, responsive single-page personal portfolio to showcase projects, technical skills, achievements, and academic background. Features an interactive canvas particle background, custom text-typing animations, and a custom light/dark mode system with layout state memory using browser localStorage.	
Technologies: HTML5, CSS3, Vanilla JavaScript (ES6+), Canvas API, Lucide Icons, HSL Color Design	

LEADERSHIP & INVOLVEMENT

Orchestra Member — Sakura 2025	<i>April 2025</i>
Organizing Committee Member — CSE Hit the Grounds 2025	<i>December 2025</i>
Organizing Committee Member — SLIoT 2026	<i>December 2025</i>
Organizing Committee Member — IESL Robogames '26	<i>December 2025</i>

ACHIEVEMENTS

- Mahapola Higher Education Scholarship
- 3rd Place — BitCode 6.0
- Finalist — SLIoT 2025
- Kaggle ML Competitor

CERTIFICATIONS & COURSES

- Intro to Machine Learning — Kaggle
- Pandas — Kaggle
- Feature Engineering — Kaggle
- Data Cleaning — Kaggle
- Problem Solving (Solve) — HackerRank, 2025
- Python for Beginners — University of Moratuwa, 2023
- Java Essentials — ICET, 2023
- C++ Fundamentals — Programming Hub